Multiplying Polynomials and Special Products

Easy level · 15 problems · A worked example comes before each type.

Name: _____ Date: _____

Key Idea

Distribute every term to every term, then combine like terms. Two patterns save time:

$$(a+b)^2 = a^2 + 2ab + b^2 \qquad (a-b)^2 = a^2 - 2ab + b^2 \qquad (a+b)(a-b) = a^2 - b^2$$

Type 1 Multiplying Polynomials

Watch Me — Worked Example

Expand $(x+3)(x+5)$.

Multiply **every** term in the first by **every** term in the second (FOIL: First, Outer, Inner, Last):

First: $x \cdot x = x^2$ Outer: $x \cdot 5 = 5x$

Inner: $3 \cdot x = 3x$ Last: $3 \cdot 5 = 15$

Add and combine like terms: $x^2 + 5x + 3x + 15 = \boldsymbol{x^2 + 8x + 15}$

$\cdot$	x	5
x	x^2	$5x$
3	$3x$	15

box method: same four products

1. Expand: $(x+2)(x+3)$

2. Expand: $(x+4)(x-1)$

3. Expand: $(x-5)(x-2)$

4. Expand: $(2x+1)(x+3)$

5. Expand: $(x+1)(x^2+2x+3)$

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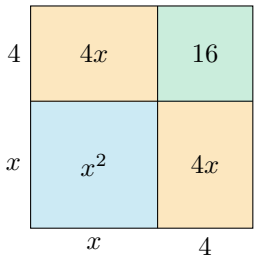
Type 2 Square of a Binomial: $(a+b)^2$ and $(a-b)^2$

Watch Me — Worked Example

Pattern: $(a+b)^2 = a^2 + 2ab + b^2$ $\qquad$ $(a-b)^2 = a^2 - 2ab + b^2$

Expand $(x+4)^2$. Here $a = x$, $b = 4$:
$x^2 + 2 \cdot x \cdot 4 + 4^2 = \boldsymbol{x^2 + 8x + 16}$

See it: a square with side $x+4$ splits into x^2, **two** rectangles of $4x$, and 16.



Expand $(x-3)^2$: $\quad x^2 - 2 \cdot x \cdot 3 + 3^2 = \boldsymbol{x^2 - 6x + 9}$

Common Mistake

$(x+4)^2 \neq x^2 + 16$. Don't forget the middle term $2ab$: $(x+4)^2 = (x+4)(x+4) = x^2 + 8x + 16$.
The last term is always **positive**: $(x-3)^2$ ends in $+9$, not -9.

6. Expand: $(x+3)^2$

7. Expand: $(x-4)^2$

8. Expand: $(x+5)^2$

9. Expand: $(2x+1)^2$

10. Expand: $(x-6)^2$

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Type 3 Sum Times Difference: $(a+b)(a-b)$

Watch Me — Worked Example

Pattern: $(a+b)(a-b) = a^2 - b^2$ ("difference of squares")

Expand $(x+6)(x-6)$.

FOIL: $x^2 - 6x + 6x - 36$. The middle terms **cancel**! $\Rightarrow$ $\boldsymbol{x^2 - 36}$

Pattern: $a = x$, $b = 6$: $x^2 - 6^2 = \boldsymbol{x^2 - 36}$

Order does not matter: $(x-6)(x+6)$ gives the same answer. *With a coefficient:* $(2x+1)(2x-1) = (2x)^2 - 1^2 = 4x^2 - 1$.

11. Expand: $(x+5)(x-5)$

12. Expand: $(x-7)(x+7)$

13. Expand: $(2x+3)(2x-3)$

14. Expand: $(x+10)(x-10)$

15. Expand: $(3x-1)(3x+1)$

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